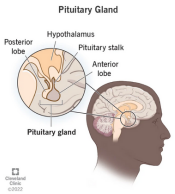
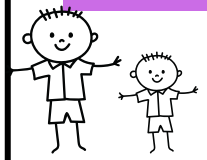
Pituitary Gland Disorders

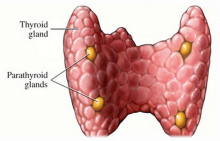


Growth Hormone Deficiency



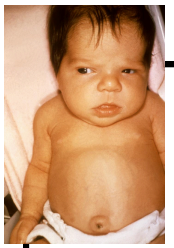
- Poor linear growth; delayed bone age.
- Plot growth velocity; endocrine testing.
- Prescribed somatropin; rotate injections.
- Monitor growth/glucose; report headache, vision change or hip pain.

Parathyroid Disorder



Hypoparathyroidism

- Low PTH → low calcium, high phosphate.
- Tingling, cramps, tetany or seizures.
- Monitor calcium, phosphate and ECG.
- Calcium + active vitamin D as ordered; urgent treatment for symptomatic hypocalcemia.

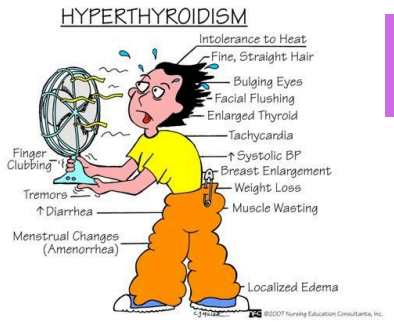


Congenital Hypothyroidism

- Often detected on newborn screen before obvious signs.
- May have prolonged jaundice, poor feeding, hypotonia, constipation.
- Confirm TSH/free T4; start levothyroxine promptly.
- Protect brain development and growth; frequent lab follow-up.



Thyroid Gland Disorders



Hyperthyroidism

- Excess thyroid hormone: heat intolerance, tachycardia, tremor, weight loss.
- Usually Graves disease in children.
- Methimazole and symptom control as ordered.
- Fever/sore throat on antithyroid drug: urgent assessment.

Metabolic Disorders

Phenylketonuria (PKU)

~~Protein~~

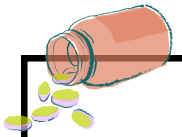


- Cannot adequately break down phenylalanine.
- Lifelong controlled phenylalanine intake + medical formula.
- Avoid aspartame; monitor blood levels.
- NOT a zero-protein diet.

Galactosemia

~~Carbs~~

- Cannot properly metabolize galactose.
- Milk exposure can cause vomiting, jaundice, poor growth or sepsis.
- Use prescribed lactose/galactose-free feeding.
- NOT avoidance of all carbohydrates.



Type 2 Diabetes

- Insulin resistance with progressive beta-cell dysfunction.
- May have polyuria, thirst or acanthosis nigricans.
- Glucose/A1c testing; evaluate symptoms urgently.
- Family-based nutrition/activity plan.
- Metformin, insulin or other age-approved medicine as prescribed.



Medications for each diagnosis?

- GH deficiency: somatropin; report severe headache or limp.
- Hypothyroidism: levothyroxine; excess causes hyperthyroid symptoms.
- Hyperthyroidism: methimazole; overtreatment can cause hypothyroidism.
- Hypoparathyroidism: calcium/calcitriol; excess can cause hypercalcemia.
- PKU: medical formula; sapropterin for eligible patients.
- Galactosemia: dietary treatment; specialist-directed supplements.
- Diabetes: excess insulin can cause hypoglycemia; monitor glucose.

Consider... what if too much of a medication is given?

Diabetes Mellitus- Type 1

Cause?

- Autoimmune beta-cell destruction → little/no insulin.
- Lifelong insulin replacement.

Assessment (s/s)

- Polyuria = much urination.
- Polydipsia = much thirst.
- Polyphagia, weight loss, fatigue.
- Vomiting + deep fruity breathing: evaluate for DKA.

Diagnosis

- Elevated blood glucose; A1c assesses chronic exposure.
- Check ketones/acidosis if ill.
- Autoantibodies help classify diabetes.

Treatment

- Basal + mealtime insulin or pump.
- Glucose/CGM monitoring, carbohydrate matching.
- Teach injection sites and hypoglycemia care.



Sick Day Rules



- Do not omit basal insulin.
- Check glucose/ketones more often; follow fluid and call instructions.



Exercise

- Check glucose; carry fast carbohydrate.
- Adjust food/insulin per plan.
- Avoid exercise with significant ketones; hypoglycemia may be delayed.



Hypoglycemia

- Sweating, shaking, hunger, confusion.
- Fast carbohydrate if safe to swallow; recheck per plan. Severe: glucagon/help.



Hyperglycemia

- Thirst, urination, fatigue.
- Check glucose/ketones; follow correction plan.

Client Teaching

- Medical ID; school/emergency plan.
- Rotate sites; safe sharps disposal.
- Know when to call for vomiting, ketones or persistent high/low glucose.

Complication: Diabetic Ketoacidosis (DKA)

- Insulin deficiency → ketones + metabolic acidosis; dehydration.
- IV fluids and insulin under pediatric protocol.
- Frequent glucose, electrolytes, neurologic and cardiac monitoring.
- Potassium may fall during treatment.
- Worsening headache/mental status: urgent escalation.